

**YOUR
HEADSHOT
HERE**

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I'm a sophomore studying physics and mathematics with a minor in data science, taking graduate-level coursework across the Claremont Colleges. I love applying mathematical tools to complex, uncertain systems — from detecting financial market crises using differential geometry to building systematic trading strategies. Advised by Prof. Trung Phan, Keck Science Department.

GitHub
QR Code

Portfolio
QR Code

I applied physics math to find hidden patterns in financial data

Can we detect market regime shifts *without labeled crisis data*?

- Embed financial time series into a **geometric space** (projective Hilbert space) using spectral metric learning

- Extract **3 geometric observables** that measure how the data manifold deforms during stress periods

- **Fully unsupervised** – no crisis labels, no regime annotations, no look-ahead bias

“QCML” is a historical name — the math is differential geometry and spectral theory, not quantum computing.

QCML vs RF Figure — poster_qcml_vs_rf.png

- Multi-Lag Fidelity (**d=1.44**) significantly outperformed supervised Random Forest (**d=1.13**, $p=0.002$)

- Tested across **5 ETFs** and **4 out-of-sample crises**

- Solo-authored **34-page research paper** (in revision)

Python PyTorch NumPy/SciPy Optuna Polygon.io LaTeX

Developed and validated systematic equity strategies for a quantitative fund

Building robust multi-strategy portfolios that survive regime changes and transaction costs

- Designed **modular research framework** with multiple strategy sleeves and two-stage volatility targeting

- **VIX/VVIX regime diagnostics** for dynamic leverage adjustment

- **Walk-forward validation** with 30-day embargo and full robustness test suite

Quanta Metrics — poster_quanta_metrics.png

- **Sharpe 2.92** and **Calmar 5.02** on out-of-sample data (2022–2025)

- Passed **full robustness test suite** (parameter stability, bootstrap, walk-forward)

- Built **C++ Heston pricer** for options strategy component (pybind11)

Python scikit-learn Optuna PostgreSQL FRED API C++

Hard Skills

Tech Python (NumPy, Pandas, PyTorch) SQL Git/CI-CD Docker LaTeX Mathematica

CS Data Structures & Algorithms ML/PyTorch Numerical Optimization

Finance Stochastic Calculus Options Pricing (BS, Heston) Backtesting/Walk-Forward Risk Metrics (Sharpe, VaR, CVaR)

Soft Skills

CS Teaching Assistant

Held weekly office hours, mentored 60+ students in intro CS

Independent Researcher

Solo-authored 34-page research paper with 3 theorems

Team Collaboration

Student Quant Fund leadership, group strategy development

Technical Communication

Translating complex math for diverse audiences

Pitzer College | Keck Science Department

Carpe Diem Endowed Scholar

Seeking: Quant Research/Trading | ML/AI Engineering | Physics Research

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